

How Exactly Does It Work?

Let's say users A, B, C and D all want file X. In the traditional HTTP or FTP scenario, a server would host X, and A, B, C and D would all download from the server. This would result in the server's upload bandwidth being shared by all four users. BitTorrent, instead, breaks X into many little chunks and lets all four users download the chunks from the server. However, it also allows for A, B, C and D to download from each other. Owing to the fact that a standard 700 MB file will be broken into, say, 700 x 1 MB parts, user A may be downloading parts 1 through 100 from the server, 101 through 300 from B, 301 to 650 from C (who might have the fastest connection of the lot), and 651 through 700 from D. This helps A achieve a much higher speed than he could normally get from just the server, and this also saves the server from having to upload the file four times to the four different users. Remember, this illustration is simplistic; in reality, the BitTorrent network is quite a bit more complicated—A could be downloading parts 4, 203, 391, 456, 666 and 700 from the server, and a variety of combinations of parts from the others; he could be downloading nothing at all from the server, etc. It's all intelligent, actually. If, say, A's connection with C were the fastest, A would end up downloading most of the file from C.

So There's Still One Server?

Not really. The BitTorrent (BT) network functions in quite a different way. There really is no way to distinguish one peer from another, so there's no server-client relationship. Everyone is equal in BT land. The only distinction there is, which is related to the server-client relationship of HTTP, is *Seeds* and *Peers*. A Seed is a computer on the BT network that has the full file, and is still uploading it for those who want it; a Peer is a computer that's both downloading and uploading the same file. So let's say you make your own Linux Distro. Initially, you will be the only seed, and those downloading from you will be peers. Let's say that in an hour or two, three users have finished downloading your distro, and are still connected to the network—these three will also become seeds. In fact, you could even go offline; if at least one of those three (who finished downloading the file) stay online till someone else also finishes and becomes a seed, the file will continue to propagate across the Net. As long as people wait for more Peers to become Seeds before disconnecting, the file transfers never die.

Unlike Web protocols, where more users online slow down the experience for everyone, BT networks thrive on popularity. So if there are millions of seeds and peers for one file, the faster you will be able to download it.

How Do Peers Share Files?

You need a BitTorrent Client—a program that uploads and downloads using the BT network. If you wanted to distribute a file X, you would have to make a Torrent file. What the software

does is look at the file or folder that you're sharing, calculate size (for files) and the number of files (for folders), and then break the data into equally-sized, manageable chunks (or bits). It then stores all this information, including your IP address, in a file called the Torrent file. If you want to share file X with your friends, you just need to send them the Torrent file, which their BT clients will interpret—and start downloading the file (or folder) from you.

So It's All Just Private Sharing?

Not at all—otherwise BitTorrent wouldn't be as popular as it is. All it takes to share something—using BT—with the world is to give the world the Torrent file; this is easily achieved using Web protocols to host the Torrent files from millions of users on a site where they can all search for and download them. Now when you're dealing with a million peers downloading a specific file, the number of people who complete downloading and are now uploading (becoming seeds) will change dramatically. This is where the "Tracker" comes into the picture in a BT network. A tracker is just an application that monitors the current status of seeds and peers downloading a particular torrent.

To make a torrent file, you go to File > New—or the equivalent of that action—in your BT client. While making your file, you will notice that most clients, by default, offer to make the torrent "Trackerless" using the public DHT (Distributed Hash Table) network. This means that all the peers downloading that torrent will have and update the information about all the other peers and seeds. When you choose to use trackers, the DHT network is disabled. This means that those downloading the torrent connect to the tracker the first time and get information about the seeds and peers for that torrent file. The tracker usually updates itself at regular intervals. Private trackers are generally used by closed groups of file sharers, and apart from monitoring download progress and the amount of seeds and peers, they also monitor Share Ratios.

Share Ratios?

Most of the older P2P networks died because of one main reason—leechers, a.k.a. P2P vermin. The older protocols had one major flaw: they didn't force people to upload. Let's take the file X example again: you start sharing the file, people connect, they download, and once they're done, they disconnect. This results in file X not being shared with the world unless you're online and sharing it. If there was some way in which you could force all those downloading to upload at least as much as they download, you could make the P2P world a better place. Bram Cohen had exactly this in mind when he designed BitTorrent: he introduced the concept of "Share Ratio" into BT, which, simply put, is the ratio of the amount of data you've uploaded to the amount you've downloaded. And so you have two share ratios: one for each specific Torrent task, and one total of all your BT participation. The BT network also